

Honglin He

Ph.D. Student in Computer Science, University of California, Los Angeles

Vision and Autonomy Intelligence Lab (VAIL) • Advisor: [Bolei Zhou](#)

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RESEARCH INTERESTS

Embodied AI, urban simulation, and robot navigation — building scalable simulation platforms and learning-based navigation policies that help robots move safely through the sidewalks and public spaces people share every day.

EDUCATION

Sep 2025 – Present	University of California, Los Angeles (UCLA) Ph.D. in Computer Science GPA 4.0/4.0 Advisor: Bolei Zhou
Sep 2021 – Jul 2024	Tsinghua University (THU) M.S. in Control Science and Engineering GPA 4.0/4.0, Rank 1/42 Advisors: Yi Zhang , Shuo Feng
Sep 2017 – Jul 2021	Xiamen University (XMU) B.E. in Automation GPA 3.7/4.0

PUBLICATIONS

* denotes equal contribution. Full list on [Google Scholar](#).

Preprints and Under Review

arXiv 2026	Monocular 3D Occupancy Perception for Robots on Sidewalks via Hybrid 2D–3D Learning Yukai Ma, Joe Lin, Liu Liu, Honglin He , Lulu Ricketts, Brad Squicciarini, Yong Liu, Bolei Zhou project page
arXiv 2026	Slow Brain, Fast Planner: Latency-Resilient VLM-Augmented Urban Navigation Zhenghao Peng, Honglin He , Quanyi Li, Yukai Ma, Bolei Zhou paper
arXiv 2025	Dreamland: Controllable World Creation with Simulator and Generative Models Sicheng Mo, Ziyang Leng, Leon Liu, Weizhen Wang, Honglin He , Bolei Zhou project page

Conference Publications

CoRL 2026	SidewalkBench: Benchmarking Visual Navigation on Urban Sidewalks (Spotlight) Zhizheng Liu*, Honglin He* , Vivek Alumootil*, Akshat Pandya, Brad Squicciarini, Wayne Wu, Bolei Zhou project page
CoRL 2026	From Imitation to Alignment: Human-Preference Flow Policies for Long-Horizon Sidewalk Navigation Honglin He , Zhizheng Liu, Yukai Ma, Bolei Zhou project page
CoRL 2026	BridgeSim: Unveiling the OL–CL Gap in End-to-End Autonomous Driving Seth Z. Zhao, Luobin Wang, Hongwei Ruan, Yuxin Bao, Yilan Chen, Ziyang Leng, Abhijit Ravichandran, Honglin He , Zewei Zhou, Xu Han, Abhishek Peri, Zhiyu Huang, Pranav Desai, Henrik Christensen, Jiaqi Ma, Bolei Zhou project page
CVPR 2026	AURA: Multi-modal Shared Autonomy for Urban Navigation Yukai Ma, Honglin He , Selina Song, Wayne Wu, Bolei Zhou project page
ICRA 2026	Learning Sidewalk Autopilot from Multi-Scale Imitation with Corrective Behavior Expansion Honglin He , Yukai Ma, Brad Squicciarini, Wayne Wu, Bolei Zhou project page
ICLR 2026	From Seeing to Experiencing: Scaling Navigation Foundation Models with Reinforcement Learning Honglin He* , Yukai Ma*, Brad Squicciarini, Wayne Wu, Bolei Zhou project page
ICLR 2026	UrbanVerse: Scaling Urban Simulation by Watching City-Tour Videos Mingxuan Liu*, Honglin He* , Elisa Ricci, Wayne Wu, Bolei Zhou project page

CVPR 2025	Towards Autonomous Micromobility through Scalable Urban Simulation (Highlight) Honglin He*, Wayne Wu*, Todd Zhang, Jack He, Seth Zhao, Ran Gong, Quanyi Li, Bolei Zhou project page
ICLR 2025	MetaUrban: An Embodied AI Simulation Platform for Urban Micromobility (Spotlight) Honglin He*, Wayne Wu*, Jack He, Yiran Wang, Chenda Duan, Zhizheng Liu, Quanyi Li, Bolei Zhou project page
IV 2024	Few-Shot Scenario Testing for Autonomous Vehicles Based on Neighborhood Coverage and Similarity Shu Li, Jingxuan Yang, Honglin He , Yi Zhang, Jianming Hu, Shuo Feng paper
CVPRW 2024	KnowMoformer: Knowledge-Conditioned Motion Transformer for Controllable Traffic Scenario Simulation Honglin He, Shu Li, Jingxuan Yang, Linxuan He, Yi Zhang, Qiuqing Lu, Shuo Feng project page
ICCV 2023	OrthoPlanes: A Novel Representation for Better 3D-Awareness of GANs Honglin He*, Zhuoqian Yang*, Shikai Li, Bo Dai, Wayne Wu project page
ICCV 2023	DNA-Rendering: A Diverse Neural Actor Repository for High-Fidelity Human-centric Rendering Wei Cheng*, Ruixiang Chen*, Wanqi Yin*, Siming Fan*, Keyu Chen*, Honglin He , Huiwen Luo, Zhongang Cai, Jingbo Wang, Yang Gao, Zhengming Yu, Zhengyu Lin, Daxuan Ren, Lei Yang, Ziwei Liu, Chen Change Loy, Chen Qian, Wayne Wu, Dahua Lin, Bo Dai, Kwan-Yee Lin project page (#2 contributor, as listed on the project page)
ITSC 2022	Adaptive Testing for Connected and Automated Vehicles with Sparse Control Variates in Overtaking Scenarios Jingxuan Yang, Honglin He , Yi Zhang, Shuo Feng, Henry X. Liu paper

Journal Publications

T-ITS 2024	Few-Shot Testing of Autonomous Vehicles with Scenario Similarity Learning Shu Li, Honglin He , Jingxuan Yang, Jianming Hu, Yi Zhang, Shuo Feng paper
T-ITS 2023	Adaptive Safety Evaluation for Connected and Automated Vehicles with Sparse Control Variates Jingxuan Yang, Haowei Sun, Honglin He , Yi Zhang, Henry X. Liu, Shuo Feng paper

RESEARCH EXPERIENCE

Sep 2025 – Present	University of California, Los Angeles Ph.D. Student Advisor: Bolei Zhou Navigation foundation models, sidewalk autopilot, and scalable urban scene generation.	Los Angeles, CA
Apr 2024 – May 2025	University of California, Los Angeles Visiting Research Scholar Supervisor: Bolei Zhou Simulation platforms for embodied AI and autonomous micromobility.	Los Angeles, CA
Nov 2022 – May 2024	Intelligent Transportation Lab, Tsinghua University Research Student Supervisor: Shuo Feng Naturalistic driving environment reconstruction and safety evaluation for autonomous vehicles.	Beijing, China
Aug 2022 – Mar 2024	Shanghai AI Laboratory Research Intern Supervisors: Bo Dai , Wayne Wu , Kwan-Yee Lin 3D-aware generative representations and a human-centric dataset and benchmark.	Shanghai, China

AWARDS AND HONORS

2025	Amazon Trainium Fellowship
2024	Outstanding Master's Thesis, Tsinghua University
2019, 2020	First Prize Scholarship for Academic Performance, Xiamen University

ACADEMIC SERVICE

Reviewer	CVPR, ICCV, ECCV, NeurIPS, ICLR, CoRL, AAAI, ICRA, IEEE TPAMI, IEEE T-ITS
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